MD ABDULLAH AL NOMAN

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E-portfolio : https://ghosthouse26.vercel.app/

Career Objective

I aim to pursue a challenging professional opportunity where I can apply my knowledge of Computer Science, Artificial Intelligence, Data Science, and Software Development to solve real-world problems. I aspire to contribute to innovative projects, develop efficient and intelligent solutions, and continuously enhance my technical skills while adding value to the organization.

Education

American International University-Bangladesh (AIUB) — *Dhaka, Bangladesh*
Bachelor of Science in Computer Science and Engineering (CSE) — *2022–Present*

Jalalabad Cantonment Public School and College — *Sylhet, Bangladesh*
Higher Secondary Certificate (HSC); GPA: 5.00 — *2017–2019*

Jalalabad Cantonment Public School and College — *Sylhet, Bangladesh*
Secondary School Certificate (SSC); GPA: 5.00 — *2015–2017*

Experience

Siriotech — *Dhaka, Bangladesh*
Data Analyst — *Apr 2025 – Feb 2026*

- Oversaw the processing of more than 10,000 daily records using Python and SQL, achieving a 30% reduction in errors and improving data integrity for production models.
- Optimized SQL queries for dataset merging and sorting, resulting in a 20% increase in data retrieval efficiency.
- Developed dashboards to identify production bottlenecks, leading to a 15% reduction in project delays.

Gobeshona Learning Academy — *Online*
IoT & Embedded Systems Intern — *Jun 2026 – Jul 2026*

- Completed a 30-day hands-on IoT and Arduino microcontroller training.
- Built C/C++ embedded prototypes for real-time sensor data acquisition and cloud integration.
- Automated sensor-to-cloud pipelines for real-time data and remote agricultural control.

Projects

Weather Image Classification with Deep Learning **[GitHub]**

- Developed and trained a custom Convolutional Neural Network (CNN) with TensorFlow and Keras to classify meteorological images into four distinct weather categories.
- Attained a test accuracy of 92.35% through systematic model tuning, data augmentation, and dropout regularization.

122 Years of Bangladesh Climate History **[GitHub]**

- Analyzed 120+ years of global temperature and rainfall data using R (Tidyverse).
- Created descriptive statistics and visualizations with ggplot2 to reveal climate trends and seasonal variations.

Gram Dorpon – Digital Twin Village Platform **[GitHub]**

- Developed a full-stack Digital Twin platform that integrates a 2D geospatial map (Leaflet) with a 3D village simulator (React Three Fiber) for precision agriculture.
- Implemented disaster simulation logic (flood, drought, rain) to visualize environmental impacts on crops and water resources, enabling proactive rural planning.

Bangla Sentiment Pro – Bengali Sentiment & Emotion Analysis **[GitHub]**

- Created a dual-head BanglaBERT Transformer model employing PyTorch and Hugging Face Transformers for classifying Bengali sentiment and emotion at once.
- Achieved 91.2% sentiment accuracy and 87.8% accuracy for emotion classification using the proposed framework, which beats Bi-LSTM, mBERT, and single-head BanglaBERT models.

- Developed an interactive analysis dashboard utilizing Streamlit, FastAPI, and MySQL for real-time predictions, Excel processing, confidence scoring, and past business insight extraction.

TeaGuardian-AI – Tea Leaf Age & Disease Classification **[GitHub]**

- Participated in building an AI tea leaf analysis application with TensorFlow/Keras, which classifies age of tea leaves and disease detection based on the inputted images.
- Combined two pre-trained deep learning models into a web application powered by Streamlit to make predictions based on the inputted images.
- Sorted out the model utilities, mappings, and app dependencies for modular programming and made use of Git LFS to manage large files.

N-Gram Language Model with Smoothing & Interpolation **[GitHub]**

- Unigram, bigram, and trigram language models were built from scratch in Python using the book of Shakespeare "Julius Caesar" from NLTK Gutenberg corpus.
- Laplace smoothing and linear interpolation were applied to account for unknown word combinations.
- Model performance was measured using perplexity; the perplexity for linear interpolation model is 956.01 versus 3,080.00 for Laplace smoothing model. Greedy search and weighted random sampling were applied for text generation.

World News Topic Analysis – Automated Categorization Using LDA **[GitHub]**

- Scraped and preprocessed world news articles from The Daily Star using R web-scraping tools.
- Used Latent Dirichlet Allocation (LDA) topic modeling technique on the corpus of news articles for discovering latent topics including environment, energy, economics, and international relations.
- Performed exploratory data analysis and made visualizations using ggplot2 and word clouds to understand word frequencies and distribution of articles across different topics.

Hospital Management System (HMS) **[GitHub]**

- Created Hospital Management System application, which is a full stack web application developed through PHP, JavaScript, HTML, CSS, and MySQL to manage hospital administration and patient workflows.
- Created modules for the admin, doctors, and patients with authentication, profile management, doctor management, medical history management, and appointments for management functionality.
- Used PHP, JavaScript, MySQL, and JSON for scheduling management by doctors.

PHP-Based Web Application – Data Processing & Record Management **[GitHub]**

- Created a server-side web application using the languages PHP, HTML5, and CSS3 for receiving user inputs and application requests.
- Added data input, processing, update, and delete capabilities using PHP and JSON.
- Installed the application to run on an Apache web server via XAMPP environment, showcasing basic principles of server-side web programming and form handling.

Dengue Fever Prediction System – SVM-Based Classification **[GitHub]**

- Built a machine learning classification algorithm using Python and Scikit-learn, which classifies dengue positive and negative samples based on their clinical and demographic characteristics.
- Conducted categorical encoding, feature scaling, and an SVM classification algorithm utilizing the RBF kernel, along with evaluation through accuracy, precision, recall, F1 score, and confusion matrix.
- Got a test accuracy of 95.00% and mean 10-fold cross-validation accuracy of 91.50%, using Pandas, NumPy, Matplotlib, and Seaborn libraries for data handling and visualization.

Smart Food Shelf-Life Tracker **[GitHub]**

- Co-authored the development of a web application for managing food inventory and tracking food expiry dates using Java 21 and Spring Boot.
- Made use of food inventory, expiry date alert, barcode-based search, and REST API using Spring Data JPA and Hibernate.
- Employed Spring Security for authentication and built the web interface using Thymeleaf, HTML, CSS, and JavaScript.

Hand Gesture Controlled LED System **[GitHub]**

- A real-time hand gesture recognizer application was built using Python, OpenCV, MediaPipe, and cvzone that detects raised fingers via a webcam.
- Arduino UNO was programmed in Python with PyFirmata and StandardFirmata libraries for controlling five LEDs based on finger count detected by the program.
- Live hand gesture tracking visualizer and LED controller via digital pins 8–12 of Arduino.

Technical Skills

Programming Languages: Python, R, Java, C++, C, JavaScript, PHP, HTML5/CSS3
Data Science & AI: TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Plotly
Web & Frameworks: Spring Boot, React, REST APIs, FastAPI
Databases: MySQL, SQL Server, PostgreSQL
Tools: Git, GitHub, Jupyter Notebook, VS Code, MATLAB, AutoCAD

Awards

- 1st Runner-Up, Quiz Competition – Session 4 of the IEEE Education Week Talk Series, 2026.
- Best Project Idea – IEEE Young Professionals Bangladesh Networking Meet-Up, 2025.

Voluntary & Extracurricular Activities

IEEE AIUB Student Branch *Dhaka, Bangladesh*
American International University-Bangladesh *Oct 2024 – Present*

ESAB AIUB Unit Face *Dhaka, Bangladesh*
American International University-Bangladesh *Jul 2024 – Present*